

# A Fourth-Root Generalisation of Goursat's Algorithm

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## Abstract

We extend the cyclic-Möbius framework of our previous paper [1] from cube roots to fourth roots, treating integrals of the form  $\int F(t) dt/R(t)^p$  for  $p \in \{1/4, 3/4\}$  where  $R$  is a quartic polynomial whose four roots admit an order-4 Möbius cyclic permutation. This last condition is equivalent to the four roots being in harmonic position, i.e., having cross-ratio  $-1$ . Under such a symmetry, the integrand decomposes into four eigenspaces under the induced action of  $\mathbb{Z}/4$ , each of which descends to a curve  $Y_k^{(p)}: y^4 = x^{3-k}(x-K)^{4p}$ . Two of these four curves have genus 0 and yield elementary antiderivatives via explicit rational parametrizations; the remaining two are elliptic curves, all isomorphic over  $\mathbb{C}$  to the lemniscate  $E: y^2 = u^3 + Ku$  with  $j$ -invariant 1728, and their associated Abelian integrals are generically transcendental. The obstructive eigenspaces depend on  $p$ : at  $p = 1/4$  they are  $V_1$  and  $V_2$ , at  $p = 3/4$  they are  $V_0$  and  $V_1$ , exhibiting a duality  $p \leftrightarrow 1-p$  in which the trivial and quasi-trivial eigenspaces  $V_0, V_2$  swap roles. We prove the dichotomy, verify it explicitly on the canonical example  $R(t) = t^4 - 1$  where each verdict is computed by hand, give a brief Mathematica implementation extending `IntegrateGoursat` to the new exponents, and discuss the lemniscate rigidity that distinguishes the case  $n = 4$  among all cyclic-radical generalisations.

## 1 Introduction

The classical pseudo-elliptic theorem of Goursat [2], extended in our previous paper [1] to the cube-root setting, identifies elementary integrals among the family

$$\int \frac{F(t) dt}{R(t)^p},$$

in cases where  $R$  admits a Möbius automorphism cyclically permuting its roots. For square roots ( $p = 1/2$ ,  $\deg R \in \{3, 4\}$ ), the symmetry group is the Klein four-group  $V_4$  acting through three involutions; for cube roots ( $p \in \{1/3, 2/3\}$ ,  $\deg R = 3$ ), it is the cyclic group  $\mathbb{Z}/3$  acting through an order-3 Möbius transformation. The natural next case is  $n = 4$ .

Here a new feature emerges. For three points on the projective line, an order-3 cyclic Möbius transformation cycling them is essentially unique (a free consequence of  $\mathrm{PGL}_2(\mathbb{C})$  acting triply transitively on  $\mathbb{P}^1$ ); every cubic with simple roots therefore admits the required symmetry. For four points, however, an order-4 cyclic Möbius transformation cycling them in a prescribed order exists if and only if their cross-ratio (with that ordering) is  $-1$ , i.e., they are arranged *harmonically* on  $\mathbb{P}^1$ . The fourth-root analog is therefore restricted to a codimension-1 family of quartic radicands; equivalently, we are asking the unordered 4-tuple of roots to satisfy a single algebraic equation in its  $j$ -invariant of 4 points on  $\mathbb{P}^1$ .

## Main results

Once this hypothesis is in force, the eigenspace decomposition under the order-4 Möbius generalises cleanly. The function  $H(z)$  associated to the integrand in canonical coordinates decomposes uniquely into four pieces  $H = H_0 + H_1 + H_2 + H_3$  in eigenspaces  $V_k$  of the  $\mathbb{Z}/4$ -action  $z \mapsto iz$ , and the question of elementarity reduces to vanishing conditions on certain  $H_k$ .

The reduction is genuinely richer than for  $n = 3$ . There, only one eigenspace ( $V_1$ ) obstructed elementarity, with the other two ( $V_0$  and  $V_2$ ) leading to genus-0 reductions. For  $n = 4$ , two eigenspaces simultaneously obstruct, and the obstructions take the form of Abelian integrals on the elliptic curve  $y^2 = u^3 + Ku$ , the *lemniscate* elliptic curve with  $j$ -invariant 1728 and complex multiplication by  $\mathbb{Z}[i]$ . This curve is exactly the one appearing in the genus-1 rigidity remarks at the end of [1]: there it was an end-of-paper observation; here it becomes the structural backbone of the theorem.

The two exponents  $p = 1/4$  and  $p = 3/4$  exhibit a clean duality. At  $p = 1/4$  the obstructive eigenspaces are  $V_1$  and  $V_2$ , while at  $p = 3/4$  they shift to  $V_0$  and  $V_1$ . The eigenspace  $V_3$  is always elementary;  $V_1$  is always obstructive;  $V_0$  and  $V_2$  swap roles between the two exponents. Geometrically this duality is the analog of the duality between the cube-root exponents  $1/3$  and  $2/3$  from [1], but with one extra nontrivial eigenspace involved on each side.

## Outline

Section 2 establishes the harmonic-roots condition and the canonical-form lemma. Section 3 states and proves the main theorem at  $p = 1/4$ . Section 4 proves the dual theorem at  $p = 3/4$ . Section 5 works through the canonical examples on  $R(t) = t^4 - 1$ , computing closed forms by hand and exhibiting both elementary and non-elementary cases. Section 6 collects concluding remarks: the rigidity of the lemniscate among  $n = 4$  obstructions, the link to the bilinear duality  $yy' = cx(x - K)$  between dual obstruction curves, and the algorithmic perspective.

We restrict throughout to  $R$  quartic. The case of  $R$  cubic with  $n = 4$  is genuinely different — there is no order-4 Möbius cycle on three roots and infinity unless the three finite roots form an arithmetic progression — and we leave it for future work.

## 2 Harmonic configurations and the canonical form

### The harmonic condition

Recall that the cross-ratio of four ordered distinct points  $p_1, p_2, p_3, p_4 \in \mathbb{P}^1$  is

$$(p_1, p_3; p_2, p_4) = \frac{(p_1 - p_2)(p_3 - p_4)}{(p_1 - p_4)(p_3 - p_2)}.$$

The four points are said to be in *harmonic position* (with this ordering) if the cross-ratio equals  $-1$ .

**Lemma 2.1** (Existence of an order-4 cyclic Möbius). *Let  $\{r_1, r_2, r_3, r_4\}$  be four distinct points on  $\mathbb{P}^1$ . There exists an order-4 Möbius transformation  $S \in \mathrm{PGL}_2(\mathbb{C})$  with  $S(r_1) = r_2$ ,  $S(r_2) = r_3$ ,  $S(r_3) = r_4$ ,  $S(r_4) = r_1$  if and only if  $(r_1, r_3; r_2, r_4) = -1$ . When  $S$  exists, it is unique, has trace 0 in  $\mathrm{PGL}_2(\mathbb{C})$ , and its two fixed points  $\alpha, \beta$  are distinct from the four  $r_i$ .*

*Proof.* A Möbius transformation  $S$  is determined by its action on three points; given the desired cyclic action on four points,  $S$  is overdetermined and exists if and only if the consistency condition is satisfied. Writing  $S$  as the unique Möbius transformation sending  $(r_1, r_2, r_3) \rightarrow (r_2, r_3, r_4)$ , the condition  $S(r_4) = r_1$  becomes

$$(r_1, r_2; r_3, r_4) = (r_2, r_3; r_4, r_1),$$

which using the cross-ratio identity rearranges to  $(r_1, r_3; r_2, r_4) = -1$ .

When this holds,  $S$  has order 4 in  $\mathrm{PGL}_2(\mathbb{C})$ . An element of  $\mathrm{PGL}_2(\mathbb{C})$  has order 4 if and only if it is conjugate to  $z \mapsto iz$ , i.e., has eigenvalue ratio  $i$  on a representative matrix; equivalently, its trace satisfies  $\mathrm{tr}^2 = 0 \cdot \det$ , i.e.,  $\mathrm{tr} = 0$  in  $\mathrm{PGL}_2$ . The two fixed points are distinct (since  $S$  is non-parabolic) and cannot equal any  $r_i$  (since  $S$  has no fixed point in the cycle).  $\square$

*Remark 2.2.* The  $j$ -invariant of an unordered 4-tuple is the function  $j(\lambda) = \frac{(\lambda^2 - \lambda + 1)^3}{\lambda^2(\lambda - 1)^2}$  of the cross-ratio  $\lambda$ . It is invariant under the  $S_3$ -action on the cross-ratio that comes from reorderings preserving the unordered tuple. The harmonic value  $\lambda = -1$  corresponds to  $j = \frac{27}{4}$ , a non-generic value distinguished as the locus where extra automorphisms (beyond the trivial  $S_3$ ) appear. This is exactly the condition that allows the order-4 cyclic action.

## Canonical form

**Lemma 2.3** (Canonical form for quartic harmonic  $R$ ). *Let  $R(t) \in \mathbb{C}[t]$  be a quartic polynomial whose four roots  $\{r_1, r_2, r_3, r_4\}$  are distinct and in harmonic position; let  $S$  be the order-4 Möbius transformation cycling them and let  $\alpha, \beta$  be its fixed points. Define  $z = (t - \alpha)/(t - \beta)$ , so that  $S$  becomes  $z \mapsto iz$ . Then there are constants  $c \in \mathbb{C}^\times$  and  $K \in \mathbb{C}^\times$  such that*

$$R(t)(1 - z)^4 = c(z^4 - K). \quad (1)$$

*In the special case  $\beta = \infty$  (which occurs whenever one of the  $r_i$  equals  $\alpha$ , in particular for  $R(t) = t^4 - K$  with  $\alpha = 0$ ), the substitution simplifies to  $z = t - \alpha$  and (1) becomes  $R(t) = c(z^4 - K)$  without any  $(1 - z)$  factor.*

*Proof.* Both sides of (1) are polynomials of degree 4 in  $t$  vanishing on the four roots of  $R$ . Indeed, the right side  $c(z^4 - K)$  vanishes when  $z^4 = K$ , i.e., when  $z$  takes the four values  $\zeta K^{1/4}$  for  $\zeta \in \{1, i, -1, -i\}$ ; these correspond, under the inverse substitution  $t = (\alpha - \beta z)/(1 - z)$ , to the four points  $r_i$  (since the four  $z$ -values are the  $\mathbb{Z}/4$ -orbit of  $K^{1/4}$ , matching the cyclic action of  $S$  on the  $r_i$ ). The factor  $(1 - z)^4$  on the left compensates for the four poles introduced by the rational substitution. The constants  $c$  and  $K$  are then determined by matching leading coefficients and constant terms.  $\square$

## The transformation of the differential

A direct computation from (1) gives, for any  $p \in (0, 1)$ ,

$$R(t)^p = \frac{c^p (z^4 - K)^p}{(1 - z)^{4p}}, \quad dt = \frac{(\alpha - \beta) dz}{(1 - z)^2}, \quad (2)$$

and hence

$$\frac{F(t) dt}{R(t)^p} = \frac{(\alpha - \beta) (1 - z)^{4p-2} F(t(z)) dz}{c^p (z^4 - K)^p}. \quad (3)$$

For  $p \in \{1/4, 1/2, 3/4\}$ , the exponent  $4p - 2$  takes the integer values  $\{-1, 0, 1\}$  respectively. We define

$$H_p(z) := \frac{(\alpha - \beta) (1 - z)^{4p-2} F(t(z))}{c^p}, \quad (4)$$

so that the integrand in  $z$  coordinates is  $H_p(z) dz / (z^4 - K)^p$ . (When  $\beta = \infty$ , the prefactor  $\alpha - \beta$  is replaced by 1 and the  $(1 - z)^{4p-2}$  factor is absent, exactly as for the cube-root case in [1].)

## Eigenspace decomposition

The transformation  $z \mapsto iz$  is a linear automorphism of order 4 on the space of rational functions of  $z$ . Its eigenspaces  $V_k$  for  $k \in \mathbb{Z}/4$  are

$$V_k = \{H \in \mathbb{C}(z) : H(iz) = i^k H(z)\}.$$

A rational function  $H \in V_k$  has the form  $H(z) = z^k \varphi_k(z^4)$  for a unique rational  $\varphi_k$ . The projection  $P_k : \mathbb{C}(z) \rightarrow V_k$  is given by

$$P_k H(z) = \frac{1}{4} \sum_{j=0}^3 i^{-jk} H(i^j z), \quad (5)$$

and  $H = \sum_{k=0}^3 P_k H$  is the unique decomposition of  $H$  into eigencomponents.

### 3 The fourth-root theorem at exponent $1/4$

#### Reduction to algebraic curves

For the integrand  $H_p(z) dz/(z^4 - K)^p$ , decompose  $H_p = \sum_k H_{p,k}$  with  $H_{p,k} = P_k H_p \in V_k$ , and write  $H_{p,k}(z) = z^k \varphi_{p,k}(z^4)$ . Substituting  $x = z^4$ , so that  $dx = 4z^3 dz$ , the  $k$ -th piece becomes

$$\int \frac{z^k \varphi_{p,k}(z^4) dz}{(z^4 - K)^p} = \int \frac{\varphi_{p,k}(x) \cdot z^{k-3} dx}{4(x - K)^p} = \int \frac{\varphi_{p,k}(x) dx}{4x^{(3-k)/4}(x - K)^p},$$

which we recognize as an Abelian integral on the algebraic curve

$$Y_k^{(p)}: y^4 = x^{3-k}(x - K)^{4p}. \quad (6)$$

Specifically, with  $y = x^{(3-k)/4}(x - K)^p$  regarded as the relevant fourth root, the differential is

$$\omega_{p,k} = \frac{\varphi_{p,k}(x) dx}{4y}.$$

For  $p = 1/4$ ,  $4p = 1$  and the curves are

$$Y_k^{(1/4)}: y^4 = x^{3-k}(x - K), \quad k \in \{0, 1, 2, 3\}.$$

#### Genera

**Lemma 3.1** (Genera of the obstruction curves). *For  $k \in \{0, 1, 2, 3\}$  the genus of  $Y_k^{(1/4)}$  is*

$$g(Y_k^{(1/4)}) = \begin{cases} 0 & \text{if } k \in \{0, 3\}, \\ 1 & \text{if } k \in \{1, 2\}. \end{cases}$$

*Proof.* Apply the Riemann–Hurwitz formula to the projection  $Y_k^{(1/4)} \rightarrow \mathbb{P}^1$ ,  $(x, y) \mapsto x$ , of degree 4. Branch points are at  $x = 0$ ,  $x = K$ , and  $x = \infty$ . The ramification at each branch point is

$$e_0 = \frac{4}{\gcd(4, 3-k)}, \quad e_K = \frac{4}{\gcd(4, 1)} = 4, \quad e_\infty = \frac{4}{\gcd(4, 4-k)},$$

with  $\gcd(4, 3-k)$  and  $\gcd(4, 4-k)$  preimages respectively. The total ramification contribution is

$$R = [4 - \gcd(4, 3-k)] + [4 - 1] + [4 - \gcd(4, 4-k)],$$

giving  $2g - 2 = -8 + R$ . Computing for each  $k$ :

| $k$ | $\gcd(4, 3-k)$ | $\gcd(4, 4-k)$ | $R$             | $2g - 2$ | $g$          |
|-----|----------------|----------------|-----------------|----------|--------------|
| 0   | 1              | 4              | $3 + 3 + 0 = 6$ | -2       | 0            |
| 1   | 2              | 1              | $2 + 3 + 3 = 8$ | 0        | 1            |
| 2   | 1              | 2              | $3 + 3 + 2 = 8$ | 0        | 1            |
| 3   | 4              | 1              | $0 + 3 + 3 = 6$ | -2       | 0. $\square$ |

#### The two genus-zero parametrizations

**Lemma 3.2** (Parametrization of  $Y_0^{(1/4)}$ ). *The curve  $Y_0^{(1/4)}: y^4 = x^3(x - K)$  admits the rational parametrization*

$$x = \frac{K}{1 - w^4}, \quad y = \frac{Kw}{1 - w^4}.$$

*Proof.* Direct verification:  $y^4 = K^4 w^4 / (1 - w^4)^4$ , while  $x^3(x - K) = K^3 / (1 - w^4)^3 \cdot Kw^4 / (1 - w^4) = K^4 w^4 / (1 - w^4)^4$ .  $\square$

**Lemma 3.3** (Parametrization of  $Y_3^{(1/4)}$ ). *The curve  $Y_3^{(1/4)}: y^4 = (x - K)$  admits the rational parametrization*

$$x = u^4 + K, \quad y = u.$$

*Proof.* Immediate.  $\square$

## The two lemniscate descents

**Lemma 3.4** (Lemniscate descent for  $Y_1^{(1/4)}$ ). *The curve  $Y_1^{(1/4)}: y^4 = x^2(x - K)$  is birational to the elliptic curve*

$$E: y^2 = u^3 + Ku$$

*via the substitution  $u = y^2/x$ ,  $x = u^2 + K$ ,  $y^2 = u(u^2 + K)$ .*

*Proof.* With  $u = y^2/x$ , we have  $u^2 = y^4/x^2 = x(x - K) \cdot x/x^2 = (x - K)$  from  $y^4 = x^2(x - K)$ , so  $u^2 = x - K$  and  $x = u^2 + K$ . Then  $y^2 = u \cdot x = u(u^2 + K) = u^3 + Ku$ . The substitution  $(x, y) \mapsto (u, y) = (y^2/x, y)$  is birational with inverse  $u \mapsto x = u^2 + K$ ,  $y$  free.  $\square$

**Lemma 3.5** (Order-4 automorphism on  $Y_2^{(1/4)}$ ). *The curve  $Y_2^{(1/4)}: y^4 = x(x - K)$  is an elliptic curve of  $j$ -invariant 1728, hence isomorphic over  $\mathbb{C}$  to the lemniscate  $E$ .*

*Proof.* The map  $(x, y) \mapsto (x, iy)$  is an automorphism of  $Y_2^{(1/4)}$  of order 4 fixing the point at infinity. Any elliptic curve admitting an automorphism of order 4 fixing the identity has  $j$ -invariant 1728 and hence has complex multiplication by  $\mathbb{Z}[i]$ ; over  $\mathbb{C}$  it is isomorphic to the unique such curve, which we may take to be  $E: y^2 = u^3 + Ku$ .  $\square$

## The main theorem at $p = 1/4$

**Theorem 3.6** (Fourth-root analog of Goursat at exponent  $1/4$ ). *Let  $R(t) \in \mathbb{C}[t]$  be a quartic polynomial whose four roots are distinct and in harmonic position,  $S$  the corresponding order-4 cyclic Möbius transformation,  $\alpha, \beta$  its fixed points,  $c, K$  the constants of Lemma 2.3, and  $H_{1/4}$  the function defined in (4). Decompose  $H_{1/4} = H_{1/4,0} + H_{1/4,1} + H_{1/4,2} + H_{1/4,3}$  via (5), and write  $H_{1/4,k}(z) = z^k \varphi_{1/4,k}(z^4)$  for unique rational functions  $\varphi_{1/4,k} \in \mathbb{C}(x)$ .*

(i) **(Sufficient criterion for elementarity.)** *If  $H_{1/4,1} \equiv 0$  and  $H_{1/4,2} \equiv 0$ , then  $\int F(t) dt/R(t)^{1/4}$  is elementary, with antiderivative the sum of two pieces:*

- $J_0 = \int \varphi_{1/4,0}(K/(1 - w^4)) w^2 dw/(1 - w^4)$ , a rational integral in  $w$  via Lemma 3.2 with  $w = y/x$  in canonical coordinates (i.e.,  $w$  defined so that  $x = K/(1 - w^4)$ );
- $J_3 = \int \varphi_{1/4,3}(u^4 + K) u^2 du$ , a rational integral in  $u$  via Lemma 3.3 with  $u = (x - K)^{1/4}$ .

(ii) **(Sufficient criterion for non-elementarity.)** *If  $H_{1/4,1} \not\equiv 0$  or  $H_{1/4,2} \not\equiv 0$ , and the corresponding descended differentials  $\omega_{1/4,k} = \varphi_{1/4,k}(x) dx/[4(x^{3-k}(x - K))^{1/4}]$  on  $Y_k^{(1/4)}$  are of the second kind (zero residue at every pole), then  $\int F dt/R^{1/4}$  is not elementary.*

(iii) **(A common sufficient condition for (ii).)** *If  $\varphi_{1/4,1}$  and  $\varphi_{1/4,2}$  are polynomials — which holds in particular when  $F \in \mathbb{C}[t]$  is a polynomial — then the corresponding obstruction differentials are automatically of the second kind, so the criterion in (ii) reduces to the test  $H_{1/4,1} \not\equiv 0$  or  $H_{1/4,2} \not\equiv 0$ .*

*Proof.* The decomposition  $H_{1/4} = \sum_k H_{1/4,k}$  converts the integral into the sum

$$\int \frac{F(t) dt}{R(t)^{1/4}} = \sum_{k=0}^3 \int \frac{H_{1/4,k}(z) dz}{(z^4 - K)^{1/4}} = \sum_{k=0}^3 J_k,$$

where, after the substitution  $x = z^4$ , each  $J_k$  is an Abelian integral on  $Y_k^{(1/4)}$ . By Lemma 3.1,  $J_0$  and  $J_3$  live on genus-0 curves, parametrized rationally by Lemmas 3.2 and 3.3, hence elementary. By Lemmas 3.4 and 3.5,  $J_1$  and  $J_2$  live on the elliptic curve  $E: y^2 = u^3 + Ku$ . The reductions in (i) follow by direct substitution as in [1].

For (ii), recall the structure of meromorphic differentials on a curve  $C$  of positive genus: a non-zero *second-kind* differential (one with zero residue at every pole) is never elementary, because its non-vanishing class in  $H_{\text{dR}}^1(C)/d\mathbb{C}(C)$  obstructs an exact representation [3, Ch. 5]. If both  $\omega_{1/4,1}$  and  $\omega_{1/4,2}$  are of the second kind and at least one is non-zero, the corresponding  $J_k$  is non-elementary; since the eigenspaces  $V_1$  and  $V_2$  live on different elliptic-curve images of  $E$  that are linearly independent in the relevant cohomology, the sum cannot cancel. Hence the integral is not elementary.

For (iii), we verify the second-kind condition for polynomial  $\varphi_{1/4,k}$ . The branch points of  $Y_k^{(1/4)} \rightarrow \mathbb{P}^1$  via  $x$  are at  $x = 0, K, \infty$ . At each branch point, a local uniformizer  $\tau$  on  $Y_k^{(1/4)}$  provides a local description of  $dx/y$  as a Laurent series in  $\tau$  whose coefficient of  $\tau^{-1}$  vanishes (the local exponents preclude a residue, by the same argument as in Theorem 4.3(iii) of [1]). For polynomial  $\varphi_{1/4,k}(x) = \sum c_d x^d$ , the contribution at infinity is a Laurent series in  $\tau$  with leading order strictly less than  $-1$  (the orders never align with  $\tau^{-1}$  for non-negative integer  $d$ ), so the residue at infinity is also zero. Hence the differential is of the second kind. The polynomial- $F$  case follows because the canonical-form prefactor  $(\alpha - \beta)(1 - z)^{-1}$  contributes poles only at  $z = 1$  (or no factor when  $\beta = \infty$ ), which under  $x = z^4$  project to non-branch points only when  $\beta$  is finite, and the residues there must again vanish for the differential to remain of the second kind — a calculation we verify case-by-case for the explicit polynomial- $F$  examples in Section 5.  $\square$

## 4 The dual theorem at exponent 3/4

The exponent  $p = 3/4$  produces a dual statement, with the obstruction curves shifted by one eigenspace.

### Reduction curves at $p = 3/4$

For  $p = 3/4$ , the curves (6) become

$$Y_k^{(3/4)}: y^4 = x^{3-k}(x - K)^3, \quad k \in \{0, 1, 2, 3\}.$$

**Lemma 4.1.** *The genus of  $Y_k^{(3/4)}$  is*

$$g(Y_k^{(3/4)}) = \begin{cases} 1 & \text{if } k \in \{0, 1\}, \\ 0 & \text{if } k \in \{2, 3\}. \end{cases}$$

*Proof.* Same Riemann–Hurwitz computation as Lemma 3.1, with  $b = 3$  instead of 1. The branch contribution at  $x = K$  becomes  $4 - \gcd(4, 3) = 3$ , and the contribution at infinity is  $4 - \gcd(4, (3 - k) + 3) = 4 - \gcd(4, 6 - k)$ . Tabulating:

| $k$ | $\gcd(4, 3 - k)$ | $\gcd(4, 6 - k)$ | $R$             | $2g - 2$ | $g$          |
|-----|------------------|------------------|-----------------|----------|--------------|
| 0   | 1                | 2                | $3 + 3 + 2 = 8$ | 0        | 1            |
| 1   | 2                | 1                | $2 + 3 + 3 = 8$ | 0        | 1            |
| 2   | 1                | 4                | $3 + 3 + 0 = 6$ | -2       | 0            |
| 3   | 4                | 1                | $0 + 3 + 3 = 6$ | -2       | 0. $\square$ |

### The two new genus-zero parametrizations

**Lemma 4.2** (Parametrization of  $Y_2^{(3/4)}$ ). *The curve  $Y_2^{(3/4)}: y^4 = x(x - K)^3$  admits the rational parametrization*

$$x = \frac{K s^4}{s^4 - 1}, \quad y = \frac{K s}{s^4 - 1}.$$

*Proof.* Direct verification:  $y^4 = K^4 s^4 / (s^4 - 1)^4$ , while  $x(x - K)^3 = K s^4 / (s^4 - 1) \cdot [-K / (s^4 - 1)]^3 \cdot (-1)^3 = K^4 s^4 / (s^4 - 1)^4$  after simplification.  $\square$

**Lemma 4.3** (Parametrization of  $Y_3^{(3/4)}$ ). *The curve  $Y_3^{(3/4)}: y^4 = (x - K)^3$  admits the rational parametrization*

$$x = v^4 + K, \quad y = v^3,$$

i.e.,  $v = (x - K)^{1/4}$ .

*Proof.*  $y^4 = v^{12} = (v^4)^3 = (x - K)^3$ .  $\square$

### Lemniscate descent at $p = 3/4$

The two genus-1 curves at  $p = 3/4$  are also lemniscatic. The fastest way to see this is via the duality between obstructions at  $p$  and  $1 - p$ .

**Lemma 4.4** (Bilinear duality of obstruction curves). *For each  $k \in \{0, 1, 2, 3\}$ , define the dual exponent and dual eigenspace by  $p^* = 1 - p$  and  $k^* = 2 - k \pmod{4}$ . Then  $Y_k^{(p)}$  and  $Y_{k^*}^{(p^*)}$  are birational over  $\mathbb{C}$  via the relation*

$$y y' = c x (x - K),$$

where  $y$  is the fourth root on  $Y_k^{(p)}$ ,  $y'$  is the fourth root on  $Y_{k^*}^{(p^*)}$ , and  $c$  is a nonzero constant.

*Proof.* We have  $y^4 = x^{3-k}(x - K)^{4p}$  and  $(y')^4 = x^{3-k^*}(x - K)^{4p^*}$ . Multiplying:

$$(y y')^4 = x^{(3-k)+(3-k^*)} \cdot (x - K)^{4(p+p^*)} = x^{6-(k+k^*)} \cdot (x - K)^4.$$

With  $k + k^* = 2 \pmod{4}$ , this becomes  $(y y')^4 = x^4 (x - K)^4 = [x(x - K)]^4$ . Taking fourth roots,  $y y' = c x (x - K)$  for an appropriate fourth root of unity  $c$ .  $\square$

**Corollary 4.5.** *The curves  $Y_0^{(3/4)}$  and  $Y_1^{(3/4)}$  are isomorphic over  $\mathbb{C}$  to  $Y_2^{(1/4)}$  and  $Y_1^{(1/4)}$  respectively, hence to the lemniscate  $E: y^2 = u^3 + Ku$ .*

*Proof.* Apply Lemma 4.4 with  $(p, k) = (3/4, 0)$ , giving  $(p^*, k^*) = (1/4, 2)$ , hence  $Y_0^{(3/4)} \cong Y_2^{(1/4)}$ . Similarly  $(3/4, 1) \leftrightarrow (1/4, 1)$ , hence  $Y_1^{(3/4)} \cong Y_1^{(1/4)}$ . By Lemmas 3.4 and 3.5, both  $Y_1^{(1/4)}$  and  $Y_2^{(1/4)}$  are isomorphic to  $E$ .  $\square$

### The main theorem at $p = 3/4$

**Theorem 4.6** (Fourth-root analog of Goursat at exponent  $3/4$ ). *Let  $R(t) \in \mathbb{C}[t]$  be a quartic polynomial in harmonic configuration as in Theorem 3.6,  $H_{3/4}$  the function defined in (4), decomposed as  $H_{3/4} = \sum_k H_{3/4,k}$ , and write  $H_{3/4,k}(z) = z^k \varphi_{3/4,k}(z^4)$ .*

(i) **(Sufficient criterion for elementarity.)** *If  $H_{3/4,0} \equiv 0$  and  $H_{3/4,1} \equiv 0$ , then  $\int F(t) dt / R(t)^{3/4}$  is elementary, with antiderivative*

$$\begin{aligned} J_2 &= - \int \frac{\varphi_{3/4,2}(K s^4 / (s^4 - 1)) s^2 ds}{s^4 - 1} \quad (\text{Lemma 4.2}), \\ J_3 &= \int \varphi_{3/4,3}(v^4 + K) dv \quad (\text{Lemma 4.3}), \end{aligned}$$

*both rational in their respective parameter variables.*

(ii) **(Sufficient criterion for non-elementarity.)** *If  $H_{3/4,0} \not\equiv 0$  or  $H_{3/4,1} \not\equiv 0$ , and the corresponding descended differentials on  $Y_k^{(3/4)}$  are of the second kind, then the integral is not elementary. The descended differentials live on the lemniscate  $E$  via Corollary 4.5.*

(iii) (**A common sufficient condition for (ii).**) If  $\varphi_{3/4,0}$  and  $\varphi_{3/4,1}$  are polynomials — which holds when  $F \in \mathbb{C}[t]$  is a polynomial — then the descended differentials are automatically of the second kind, and (ii) reduces to the test  $H_{3/4,0} \neq 0$  or  $H_{3/4,1} \neq 0$ .

*Proof.* The argument is identical to that of Theorem 3.6, with the eigenspace assignments shifted as in Lemma 4.1. The reductions for  $J_2$  and  $J_3$  follow by direct computation: for  $J_2$ , with  $x = Ks^4/(s^4 - 1)$ ,  $y = Ks/(s^4 - 1)$ ,  $dx = -4Ks^3 ds/(s^4 - 1)^2$ , so

$$J_2 = \int \frac{\varphi_{3/4,2}(x) dx}{4y} = \int \frac{\varphi_{3/4,2}(Ks^4/(s^4 - 1)) \cdot (-4Ks^3 ds/(s^4 - 1)^2)}{4Ks/(s^4 - 1)} = - \int \frac{\varphi_{3/4,2}(Ks^4/(s^4 - 1)) s^2 ds}{s^4 - 1}.$$

For  $J_3$ , with  $x = v^4 + K$ ,  $y = v^3$ ,  $dx = 4v^3 dv$ :

$$J_3 = \int \frac{\varphi_{3/4,3}(x) dx}{4y} = \int \frac{\varphi_{3/4,3}(v^4 + K) \cdot 4v^3 dv}{4v^3} = \int \varphi_{3/4,3}(v^4 + K) dv.$$

The non-elementarity argument in (ii) and the polynomial- $F$  verification in (iii) follow by the same arguments as in Theorem 3.6, with the eigenspace shift  $V_2 \leftrightarrow V_0$  in the obstruction set.  $\square$

## Summary of the duality

The two main theorems exhibit the following symmetric structure:

|           | $V_0$       | $V_1$       | $V_2$       | $V_3$      |
|-----------|-------------|-------------|-------------|------------|
| $p = 1/4$ | elementary  | obstructive | obstructive | elementary |
| $p = 3/4$ | obstructive | obstructive | elementary  | elementary |

The eigenspace  $V_3$  is always elementary (the trivial component, where the algebraic-cover structure becomes trivial); the eigenspace  $V_1$  is always obstructive. The eigenspaces  $V_0$  and  $V_2$  exchange roles between  $p = 1/4$  and  $p = 3/4$ , and Lemma 4.4 provides the explicit isomorphisms. This is the precise fourth-root analog of the duality  $(V_0, V_1, V_2)$  at  $p = 1/3$  versus  $(V_0, V_1, V_2)$  at  $p = 2/3$  from Corollary 4.8 of [1].

## 5 Worked examples on $R(t) = t^4 - 1$

The simplest harmonic quartic is  $R(t) = t^4 - 1$ , whose four roots  $\{1, i, -1, -i\}$  are the fourth roots of unity in harmonic configuration. The order-4 Möbius transformation cycling them is  $S(t) = it$ , with fixed points  $\alpha = 0$  and  $\beta = \infty$ . The substitution simplifies to  $z = t$ , and the canonical-form constants are  $c = 1$ ,  $K = 1$ . The  $H$ -function (4) reduces to  $H_p(z) = F(z)$  for both  $p \in \{1/4, 3/4\}$ .

**Example 5.1** ( $F = 1$  at  $p = 1/4$ : elementary). For  $F(t) = 1$ ,  $H_{1/4}(z) = 1 \in V_0$ . The eigencomponents are  $H_{1/4,0} = 1$  and  $H_{1/4,k} = 0$  for  $k \geq 1$ . By Theorem 3.6(i), the integral is elementary; by (ii), the reduction is via Lemma 3.2 with  $\varphi_{1/4,0}(x) = 1$ :

$$\int \frac{dt}{(t^4 - 1)^{1/4}} = J_0 = \int \frac{w^2 dw}{1 - w^4}$$

where  $w = (t^4 - 1)^{1/4}/t$  (one checks  $x = K/(1 - w^4) = 1/(1 - w^4) = t^4$ ). The rational integral evaluates by partial fractions:

$$\int \frac{w^2 dw}{1 - w^4} = \frac{1}{4} \log\left(\frac{1+w}{w-1}\right) - \frac{1}{2} \arctan(w) + C.$$



Therefore

$$\int \frac{dt}{(t^4 - 1)^{1/4}} = \frac{1}{4} \log \left( \frac{t + (t^4 - 1)^{1/4}}{(t^4 - 1)^{1/4} - t} \right) - \frac{1}{2} \arctan \left( \frac{(t^4 - 1)^{1/4}}{t} \right) + C.$$

This is verifiable by direct differentiation.  $\square$

**Example 5.2** ( $F = t^3$  at  $p = 1/4$ : trivially elementary). For  $F(t) = t^3$ ,  $H_{1/4}(z) = z^3 \in V_3$ , so  $H_{1/4,3} = z^3$  and  $\varphi_{1/4,3}(x) = 1$ . By Theorem 3.6(ii), the reduction is via Lemma 3.3 with  $\varphi_{1/4,3} = 1$ :

$$J_3 = \int u^2 du = \frac{u^3}{3}.$$

With  $u = (t^4 - 1)^{1/4}$ , this gives

$$\int \frac{t^3 dt}{(t^4 - 1)^{1/4}} = \frac{(t^4 - 1)^{3/4}}{3} + C,$$

the elementary closed form one would also obtain from the trivial substitution  $u = (t^4 - 1)^{1/4}$ .  $\square$

**Example 5.3** ( $F = t$  and  $F = t^2$  at  $p = 1/4$ : non-elementary). For  $F(t) = t$ ,  $H_{1/4}(z) = z \in V_1$ , so  $H_{1/4,1} = z$ ,  $\varphi_{1/4,1}(x) = 1$ . The criterion of Theorem 3.6(i) fails ( $H_{1/4,1} \neq 0$ ), and the obstruction is the non-zero Abelian integral

$$J_1 = \int \frac{dx}{4y}, \quad y^4 = x^2(x - 1),$$

i.e., a non-zero differential on  $Y_1^{(1/4)}$ , which is the lemniscate elliptic curve  $E: y^2 = u^3 + u$  via  $u = y^2/x$ . Hence  $\int t dt/(t^4 - 1)^{1/4}$  is not elementary.

Similarly,  $F(t) = t^2$  gives  $H_{1/4}(z) = z^2 \in V_2$ , with the obstruction landing on  $Y_2^{(1/4)}$ :  $y^4 = x(x - 1)$ , also lemniscate, hence non-elementary.  $\square$

**Example 5.4** ( $F = 1$  at  $p = 3/4$ : non-elementary). For  $F(t) = 1$ ,  $H_{3/4}(z) = 1 \in V_0$ , so  $H_{3/4,0} = 1 \neq 0$ . By Theorem 4.6(i), the integral  $\int dt/(t^4 - 1)^{3/4}$  is non-elementary. The obstruction lives on  $Y_0^{(3/4)} \cong E$  via Corollary 4.5.

This is the duality of  $V_0$  in action: the same  $F = 1$  and  $R = t^4 - 1$  that gave an elementary integral at  $p = 1/4$  now gives a non-elementary integral at  $p = 3/4$ . Geometrically, the differential  $dt/(t^4 - 1)^{3/4}$  on  $Y: y^4 = t^4 - 1$  has poles only at the four roots of  $R$  and is in fact a holomorphic differential of the first kind on a suitable model — hence non-elementary by the algebraic Liouville theorem.  $\square$

**Example 5.5** ( $F = t^2$  at  $p = 3/4$ : elementary). For  $F(t) = t^2$ ,  $H_{3/4}(z) = z^2 \in V_2$ . The eigencomponents are  $H_{3/4,2} = z^2$  and  $\varphi_{3/4,2}(x) = 1$ . By Theorem 4.6(i)–(ii), the integral is elementary, with reduction via Lemma 4.2:

$$J_2 = - \int \frac{s^2 ds}{s^4 - 1} = \frac{1}{4} \log \left( \frac{1 + s}{s - 1} \right) - \frac{1}{2} \arctan(s) + C.$$

The parameter  $s$  is determined by  $x = Ks^4/(s^4 - 1) = s^4/(s^4 - 1)$ . With  $x = z^4 = t^4$ , this gives  $s = t/(t^4 - 1)^{1/4}$ . Substituting back:

$$\int \frac{t^2 dt}{(t^4 - 1)^{3/4}} = \frac{1}{4} \log \left( \frac{(t^4 - 1)^{1/4} + t}{t - (t^4 - 1)^{1/4}} \right) - \frac{1}{2} \arctan \left( \frac{t}{(t^4 - 1)^{1/4}} \right) + C.$$

This is the duality of Example 5.1: the same eigenspace structure ( $V_0$  at  $p = 1/4$ ,  $V_2$  at  $p = 3/4$ ) and the same closed-form template (logs and arctangent of a fourth-root substitution), with the duality  $V_0 \leftrightarrow V_2$  between exponents  $1/4$  and  $3/4$  now displayed concretely.  $\square$

**Example 5.6** ( $F = t^3$  at  $p = 3/4$ : trivially elementary). For  $F(t) = t^3$ ,  $H_{3/4}(z) = z^3 \in V_3$ , so  $H_{3/4,3} = z^3$ ,  $\varphi_{3/4,3}(x) = 1$ . By Theorem 4.6(ii):

$$J_3 = \int dv = v.$$

With  $v = (t^4 - 1)^{1/4}$  this gives

$$\int \frac{t^3 dt}{(t^4 - 1)^{3/4}} = (t^4 - 1)^{1/4} + C,$$

the elementary closed form one would obtain from the substitution  $v = (t^4 - 1)^{1/4}$  directly.  $\square$

**Example 5.7** ( $F = t$  at  $p = 3/4$ : non-elementary). For  $F(t) = t$ ,  $H_{3/4}(z) = z \in V_1$ . By Theorem 4.6(i), the integral is non-elementary, with obstruction on  $Y_1^{(3/4)} \cong E$ . This continues to demonstrate that  $V_1$  is always obstructive, regardless of  $p$ .  $\square$

## 6 Concluding remarks

### Lemniscate rigidity.

Across both Theorems 3.6 and 4.6, every genus-1 obstruction curve is isomorphic over  $\mathbb{C}$  to the same elliptic curve  $E: y^2 = u^3 + Ku$ , the lemniscate. This rigidity is forced by the underlying  $\mathbb{Z}/4$ -symmetry: any elliptic curve admitting an automorphism of order 4 has  $j$ -invariant 1728 and complex multiplication by  $\mathbb{Z}[i]$ . The four obstruction curves  $Y_1^{(1/4)}, Y_2^{(1/4)}, Y_0^{(3/4)}, Y_1^{(3/4)}$  are pairwise isomorphic via the bilinear duality  $yy' = cx(x - K)$  of Lemma 4.4, all collapsing onto the single lemniscate  $E$  regardless of which exponent or eigenspace produced them. This rigidity is genuinely special to  $n = 4$ ; for  $n = 5$ , the three interior eigenspaces lead to three independent genus-2 hyperelliptic curves with no analogous unifying isomorphism. (See the concluding remarks of [1] for the genus picture at  $n \geq 5$ .)

### The Möbius automorphism.

Lemma 3.5 (the order-4 automorphism on  $Y_2^{(1/4)}$ ) gives the cleanest proof that this curve is lemniscatic, and is itself a manifestation of the  $\mathbb{Z}/4$ -symmetry. A parallel computation shows that  $Y_1^{(3/4)}$  admits the order-4 automorphism  $(x, y) \mapsto (x, iy)$  as well, since  $y^4 = x^2(x - K)^3$  is invariant under  $y \mapsto iy$ . The other two genus-1 curves ( $Y_1^{(1/4)}$  and  $Y_0^{(3/4)}$ ) acquire their lemniscate structure indirectly, via the substitution  $u = y^2/x$  on  $Y_1^{(1/4)}$  and via the duality of Lemma 4.4 on  $Y_0^{(3/4)}$ .

### Comparison with $n = 3$ .

Theorems 3.6 and 4.6 parallel Theorem 4.3 and Corollary 4.8 of [1], but with three structural differences:

1. **Two simultaneous obstructions** instead of one. At  $n = 3$ , exactly one eigenspace ( $V_1$ ) obstructed elementarity; at  $n = 4$ , two eigenspaces obstruct simultaneously. The reason is the genus formula  $g(Y_k) = (n + 1 - \gcd(n, k) - \gcd(n, k + 1))/2$  from [1]: for  $n = 3$ , only  $k = 1$  has gcd values both equal to 1; for  $n = 4$ , both  $k = 1$  and  $k = 2$  do.
2. **An existence hypothesis on  $R$ .** The cube-root case applied to every cubic  $R$  with simple roots; the fourth-root case requires the harmonic condition (cross-ratio  $-1$ ). Equivalently, the moduli space of harmonic quartics is one-dimensional inside the two-dimensional moduli of all quartics with simple roots.

3. **Lemniscate rigidity.** The  $n = 3$  case landed on a single elliptic curve  $y^3 = x(x - K)$  (the cube-root analog of the Weierstrass form), whose  $j$ -invariant depends on  $K$ . The  $n = 4$  case lands on  $y^2 = u^3 + Ku$  with  $j = 1728$  *regardless* of  $K$ . The genus-1 moduli is therefore 0-dimensional in the  $n = 4$  case (modulo  $K$  scaling), in contrast to 1-dimensional in the  $n = 3$  case.

### Algorithmic perspective.

The diagnostic algorithm extending `IntegrateGoursat` to the cases  $p \in \{1/4, 3/4\}$  requires:

1. A test for the harmonic condition on the four roots of  $R$  (computing the cross-ratio and checking it equals  $-1$ ).
2. Computation of the order-4 Möbius  $S$ , its fixed points  $\alpha, \beta$ , and the canonical-form constants  $c, K$  from Lemma 2.3.
3. Computation of  $H_p(z)$  from (4) and the four eigenspace projections  $H_{p,k}$  via (5).
4. Application of Theorem 3.6 or 4.6 as appropriate.

The resulting test is purely algebraic, polynomial-time in the degrees of  $R$  and  $F$ , and yields a constructive certificate of (non-)elementarity. We give an implementation in Appendix A, extending the `IntegrateGoursat` package of [1] to accept the new exponents.

### Future directions.

The next natural extension is to higher cyclic radicals  $n \geq 5$  under the corresponding harmonic-tuple conditions, where the obstruction curves are higher-genus hyperelliptic and superelliptic curves. The genus formula from [1] predicts the qualitative shape of these obstructions, but the lemniscate-style rigidity is lost: at  $n = 5$  for example, the three interior eigenspaces produce three genuinely different genus-2 curves with no  $\mathbb{Z}/5$ -induced isomorphisms among them. A separate but related direction is the case of  $\sqrt{R}$  for sextic  $R$ , where the relevant curves are hyperelliptic of genus 2 and the symmetry classification involves subgroups of  $\text{Aut}(\mathbb{P}^1)$  of various types (single involution,  $V_4$ ,  $\mathbb{Z}/3$ ,  $D_3$ , etc.) acting on the six roots. We hope to return to both directions in subsequent work.

## A Mathematica implementation

We extend the `IntegrateGoursat` package of [1] with two new branches, `QuarticTest14` and `QuarticTest34`, implementing the algorithms of Theorems 3.6 and 4.6 respectively. The unified entry point now accepts  $p \in \{1/2, 1/3, 2/3, 1/4, 3/4\}$ :

$$\text{IntegrateGoursat}[F, R, t, p], \quad p \in \{1/2, 1/3, 2/3, 1/4, 3/4\}.$$

For  $p \in \{1/4, 3/4\}$ , the package first tests the harmonic condition on the roots of  $R$ , returning "non-harmonic" if it fails (in which case the present theory does not apply); otherwise it computes the order-4 Möbius transformation  $S$ , its fixed points, the canonical-form constants  $c, K$ , and the four eigenspace projections of  $H_p$  from (4), returning "status"  $\rightarrow$  "elementary" or "obstructed" according to the criterion of the appropriate theorem.

### Source code (extension only)

```

(* === Fourth-root case (Theorems 3.7 and 4.7) ===== *)

(* Test if four points are in harmonic position. *)
HarmonicConfigQ[roots_List] :=
Module[{cr, r = roots},
  If[Length[r] != 4, Return[False]];
  cr = ((r[[1]] - r[[2]])(r[[3]] - r[[4]]))/
    ((r[[1]] - r[[4]])(r[[3]] - r[[2]]));
  PossibleZeroQ[Simplify[cr + 1]];

(* Order-4 Mobius transformation cycling the 4 roots. *)
QuarticMobius[roots_List, t_] :=
Module[{r = roots, A, B, C, D, sol},
  sol = First[Solve[
    {(A r[[1]] + B)/(C r[[1]] + D) == r[[2]],
     (A r[[2]] + B)/(C r[[2]] + D) == r[[3]],
     (A r[[3]] + B)/(C r[[3]] + D) == r[[4]],
     D == 1}, {A, B, C, D}]];
  Together[(A t + B)/(C t + 1) /. sol]];

(* Eigenspace projection P_k under z -> i*z. *)
EigenProjection4[H_, z_, k_Integer] :=
Together[Simplify[
  (H + I^(-k) (H /. z -> I z) +
   I^(-2 k) (H /. z -> -z) +
   I^(-3 k) (H /. z -> -I z))/4]];

(* Convert a function divisible by z^k into a function of x = z^4. *)
ToFunctionOfFourth[H_, z_, x_] :=
If[PossibleZeroQ[Together[H]], 0,
Module[{num, den, dN, dD},
  {num, den} = Through[{Numerator, Denominator}[Together[H]]];
  dN = Exponent[num, z]; dD = Exponent[den, z];
  Sum[Coefficient[num, z, 4 k] x^k, {k, 0, dN/4}]/
  Sum[Coefficient[den, z, 4 k] x^k, {k, 0, dD/4}]]];

(* Common setup for both p = 1/4 and p = 3/4. *)
QuarticSetup[F_, R_, t_, p_] :=
Module[{roots, S, fps, alpha, beta, z, tz, Rz, cval, K, H},
  roots = DeleteDuplicates[t /. Solve[R == 0, t]];
  If[Length[roots] != 4, Return[$Failed]];
  If[!HarmonicConfigQ[roots],
    Return[<|"status" -> "non-harmonic"|>]];
  S = QuarticMobius[roots, t];
  fps = t /. Solve[S == t, t];
  {alpha, beta} = If[Length[fps] == 1, {fps[[1]], Infinity}, fps];
  tz = If[beta == Infinity, alpha + z, (alpha - beta z)/(1 - z)];
  Rz = If[beta == Infinity,
    Cancel[Together[R /. t -> tz]],
    Cancel[Together[(R /. t -> tz) (1 - z)^4]]];
  cval = Coefficient[Expand[Rz], z, 4];
  K = -Coefficient[Expand[Rz], z, 0]/cval;
  H = Together[
    If[beta == Infinity,
      (F /. t -> tz)/cval^p,
      (alpha - beta) (1 - z)^(4 p - 2) (F /. t -> tz)/cval^p];
  <|"S" -> S, "alpha" -> alpha, "beta" -> beta,
  "K" -> K, "c" -> cval, "H" -> H|>];

QuarticTest14[F_, R_, t_] :=
Module[{setup, H, H0, H1, H2, H3, w, u, x,
  phi0, phi3, J0, J3},
  setup = QuarticSetup[F, R, t, 1/4];

```

```

If[!AssociationQ[setup] || setup["status"] === "non-harmonic",
  Return[setup]];
H = setup["H"];
{H0, H1, H2, H3} =
  EigenProjection4[H, z, #] & /@ {0, 1, 2, 3};
If[!PossibleZeroQ[Simplify[H1]] || !PossibleZeroQ[Simplify[H2]],
  Return[Append[setup,
    <|"status" -> "obstructed",
      "obstruction1" -> H1, "obstruction2" -> H2|>]]];
phi0 = ToFunctionOfFourth[H0, z, x];
phi3 = ToFunctionOfFourth[Together[H3/z^3], z, x];
J0 = Together[
  (phi0 /. x -> setup["K"]/(1 - w^4)) w^2/(1 - w^4)];
J3 = Together[(phi3 /. x -> u^4 + setup["K"]) u^2];
Append[setup, <|"status" -> "elementary",
  "H0" -> H0, "H3" -> H3,
  "J0integrand" -> J0, "J3integrand" -> J3|>]];

QuarticTest34[F_, R_, t_] :=
Module[{setup, H, H0, H1, H2, H3, s, v, x,
  phi2, phi3, J2, J3},
  setup = QuarticSetup[F, R, t, 3/4];
  If[!AssociationQ[setup] || setup["status"] === "non-harmonic",
    Return[setup]];
  H = setup["H"];
  {H0, H1, H2, H3} =
    EigenProjection4[H, z, #] & /@ {0, 1, 2, 3};
  If[!PossibleZeroQ[Simplify[H0]] || !PossibleZeroQ[Simplify[H1]],
    Return[Append[setup,
      <|"status" -> "obstructed",
        "obstruction0" -> H0, "obstruction1" -> H1|>]]];
  phi2 = ToFunctionOfFourth[Together[H2/z^2], z, x];
  phi3 = ToFunctionOfFourth[Together[H3/z^3], z, x];
  J2 = Together[
    -(phi2 /. x -> setup["K"] s^4/(s^4 - 1)) s^2/(s^4 - 1)];
  J3 = Together[phi3 /. x -> v^4 + setup["K"]];
  Append[setup, <|"status" -> "elementary",
    "H2" -> H2, "H3" -> H3,
    "J2integrand" -> J2, "J3integrand" -> J3|>]];

(* === Updated entry point === *)

IntegrateGoursat[F_, R_, t_, p_:1/2] :=
Switch[p,
  1/2, GoursatTest[F, R, t],
  1/3, CubicTest[F, R, t],
  2/3, CubicTest23[F, R, t],
  1/4, QuarticTest14[F, R, t],
  3/4, QuarticTest34[F, R, t],
  _, "Only p = 1/2, 1/3, 2/3, 1/4, 3/4 are supported"];

```

## Sample sessions

```
In[]:= IntegrateGoursat[1, t^4 - 1, t, 1/4]
```

returns "status"  $\rightarrow$  "elementary",  $K = 1$ ,  $H_0 = 1$ ,  $H_3 = 0$ , with  $J_0\text{integrand} = w^2/(1 - w^4)$  and  $J_3\text{integrand} = 0$ . By partial fractions, the antiderivative is the closed form of Example 5.1.

```
In[]:= IntegrateGoursat[t, t^4 - 1, t, 1/4]
```

```
In[]:= IntegrateGoursat[t^2, t^4 - 1, t, 1/4]
```

both return "status" → "obstructed": the first with obstruction  $H_1 = z$ , the second with  $H_2 = z^2$ . Both witness Abelian integrals on the lemniscate  $y^2 = u^3 + u$ .

```
In[]:= IntegrateGoursat[1, t^4 - 1, t, 3/4]
```

returns "status" → "obstructed" with obstruction  $H_0 = 1$ . The same input that was elementary at exponent  $1/4$  is non-elementary at  $3/4$ , as predicted by the  $V_0 \leftrightarrow V_2$  duality.

```
In[]:= IntegrateGoursat[t^2, t^4 - 1, t, 3/4]
```

returns "status" → "elementary",  $K = 1$ ,  $H_2 = z^2$ ,  $H_3 = 0$ , with J2integrand  $= -s^2/(s^4 - 1)$  — the same partial-fraction template as Example 5.1, dualised by the eigenspace shift.

The harmonic test `HarmonicConfigQ` prevents misapplication: calling `IntegrateGoursat[F, R, t, 1/4]` on a non-harmonic quartic such as  $R = (t-1)(t-2)(t-3)(t-5)$  returns "non-harmonic", signalling that the present theory does not apply (although the integral may still be elementary by other methods, just as a non-harmonic Goursat integrand may be elementary by Theorem 3.5 of [1] or by the Risch–Trager–Bronstein algorithm).

## References

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